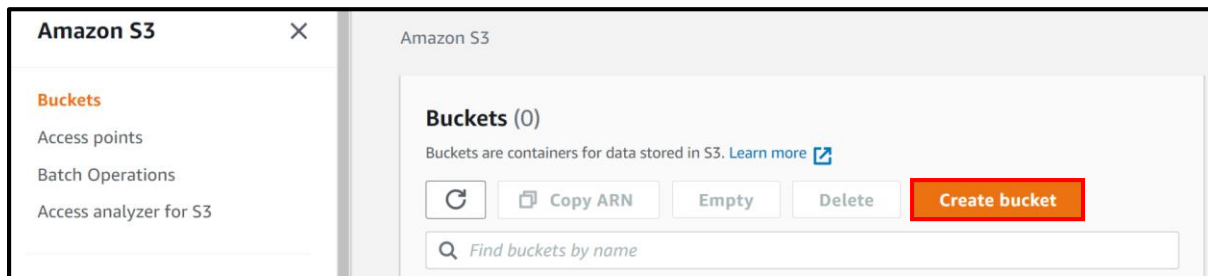


Secure AWS S3 bucket access to specific IP address.

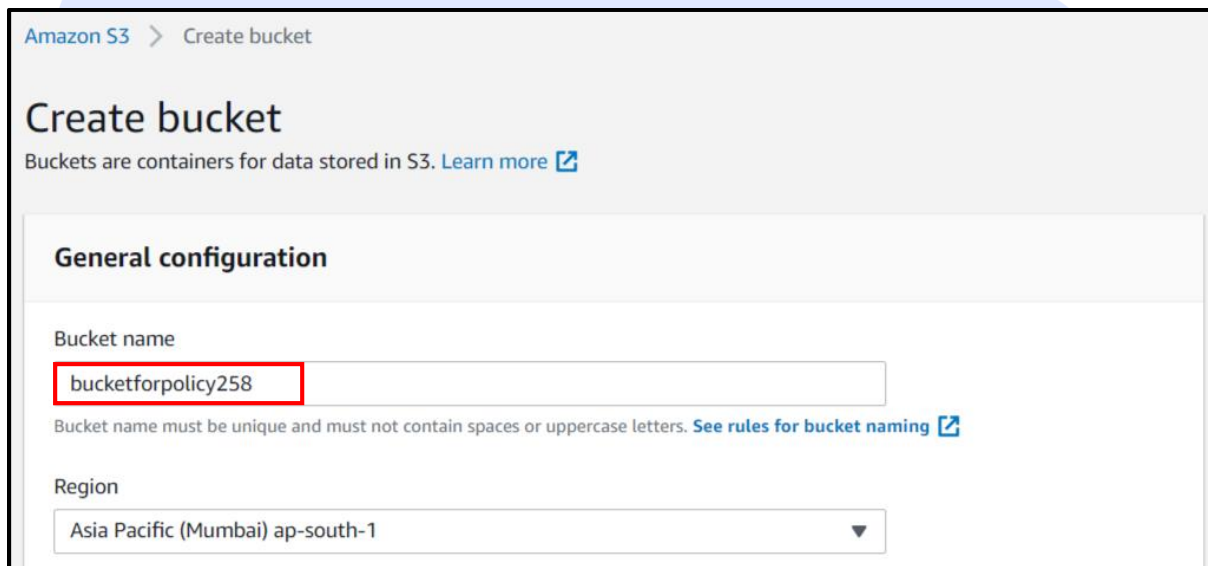
Learning Objectives:

- Ensure Bucket security by allowing only known specific address to access the contents. This can be extended to allow specific range of IP addresses.
- Learn application of resource based policies with conditional access.

Step 1: In **AWS Console** go to **S3** service. Click on **Create bucket**.



In the new window under **General Configuration** provide **Bucket name**. We use **bucketforpolicy258** as bucket name for this document.



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Scroll down and **uncheck** **Block all public access**.

Check the **acknowledgement** of these settings.

Bucket settings for Block Public Access

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

 **Turning off block all public access might result in this bucket and the objects within becoming public**
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Scroll to end of page and click on **Create bucket**.

After creating the bucket you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel Create bucket

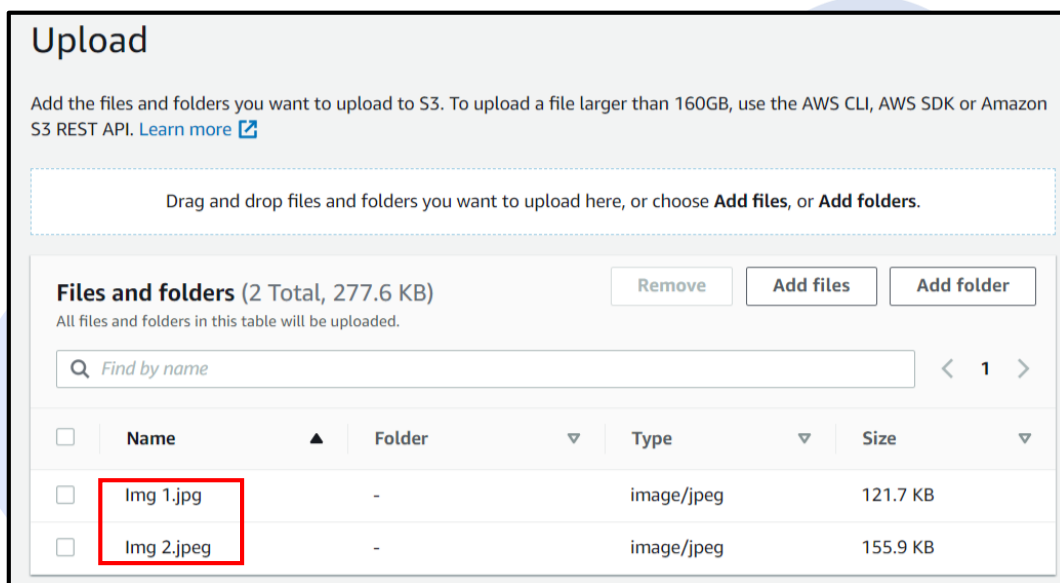
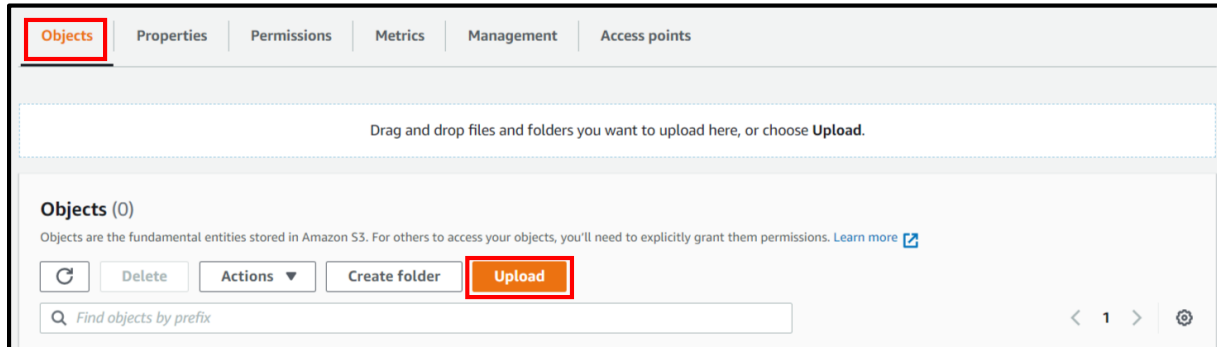
Check the bucket is created.

	Name	Region	Access
☐	bucketforpolicy258	Asia Pacific (Mumbai) ap-south-1	Objects can be public

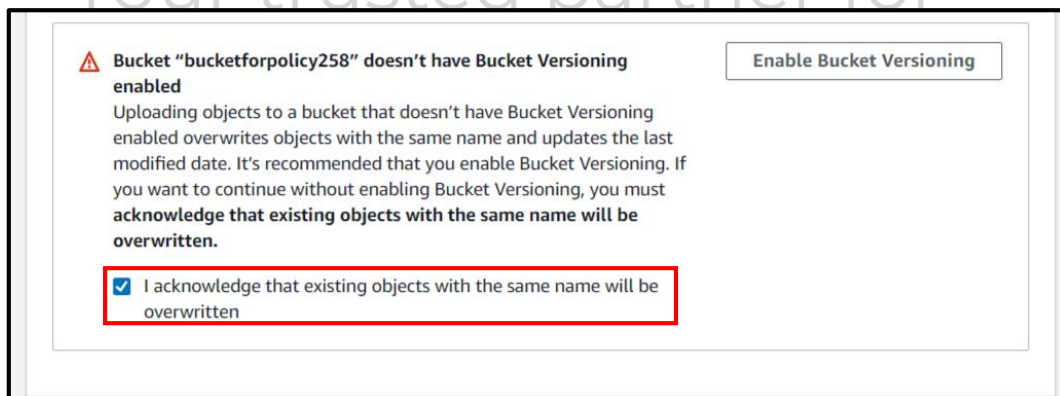
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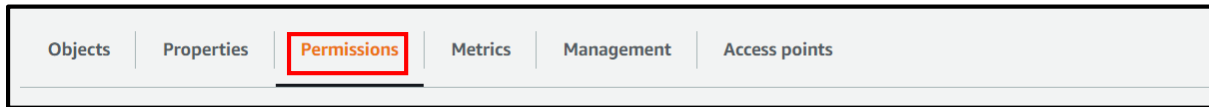
Step 2: Click on the bucket and go to **Objects** Tab. Click on **Add Files** and add 2 images to the bucket.



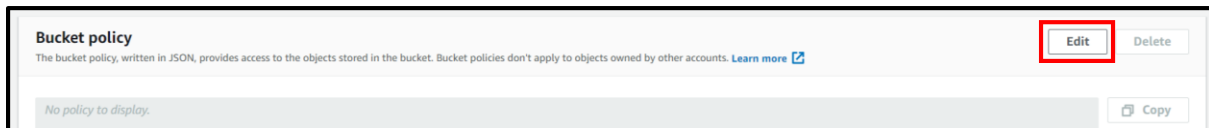
Acknowledge the disabled versioning and click on **Upload** at the end of page.



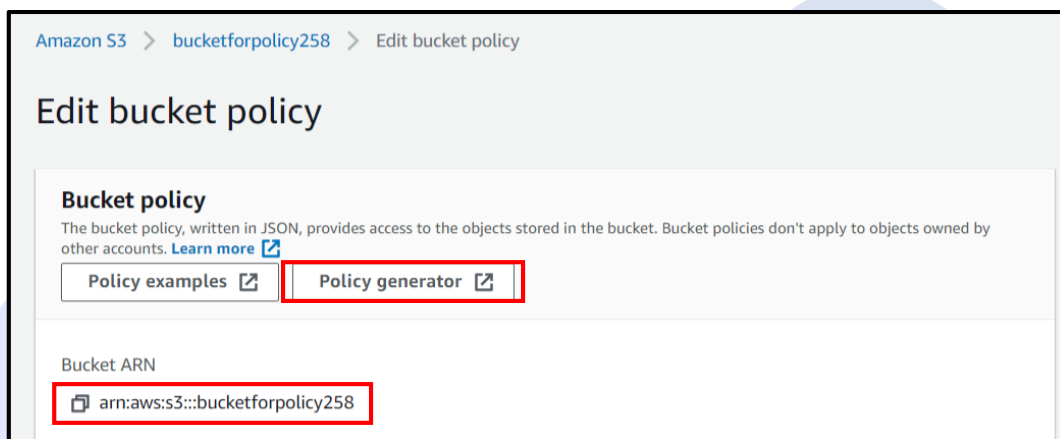
Step 3: Go back to bucket level and click on **Permissions** Tab.



Scroll down to **Bucket policy** and click on **Edit**.



In the new window first copy the **Bucket ARN** and then click on **Policy Generator**.



In **AWS Policy Generator** provide the following values:

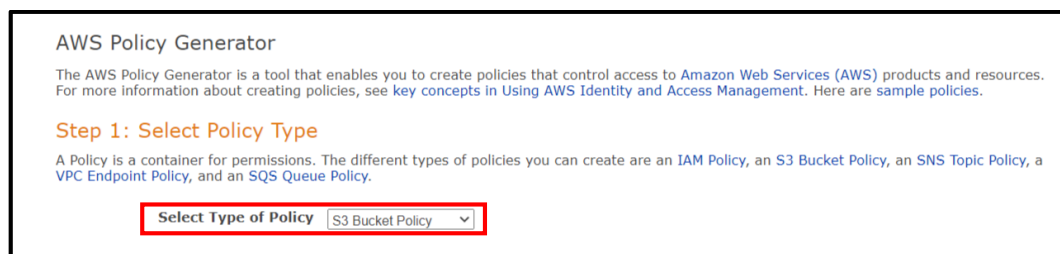
Select type of Policy: S3 Bucket Policy

Effect: Allow

Principal: * (asterix that signifies all)

Actions: **GetObject**

ARN: **arn:aws:s3:::bucketforpolicy258/*** (replace this with your copied arn along with /*)



Step 2: Add Statement(s)

A statement is the formal description of a single permission. See a [description of elements](#) that you can use in statements.

Effect ☒ Allow ☐ Deny

Principal

Use a comma to separate multiple values.

AWS Service ☐ All Services ('*')

Use multiple statements to add permissions for more than one service.

Actions ☐ All Actions ('*')

Amazon Resource Name (ARN)

☐ GetInventoryConfiguration
☐ GetJobTagging
☐ GetLifecycleConfiguration
☐ GetMetricsConfiguration
☒ **GetObject**
☐ GetObjectAcl
☐ GetObjectLegalHold
☐ GetObjectRetention

ARN should follow the following format: arn:aws:s3:::<bucket_name>/<key_name>.
Use a comma to separate multiple values.

[Add Conditions \(Optional\)](#)

Click on Add Conditions(Optional) and provide following values:

Condition: **IpAddress**

Key: **aws:SourceIp**

Value: (your **IP** address)

In order to know your IP address use the following URL:

<https://www.whatismyip.com/>

Click on **Add Condition**.

Click on **Add Statement**.

Add Conditions (Optional) Hide

Conditions are any restrictions or details about the statement. [\(More Details\)](#).

Condition

Key

Value

Condition	Keys
IpAddress	• aws:SourceIp: " "

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Scroll down and click on **Generate Policy**.

Copy the policy from the **Policy JSON Document** pop-up window.

You added the following statements. Click the button below to Generate a policy.

Principal(s)	Effect	Action	Resource	Conditions
* *	Allow	s3:GetObject	arn:aws:s3:::bucketforpolicy258/*	IpAddress aws:SourceIp: [redacted]

Step 3: Generate Policy

A policy is a document (written in the [Access Policy Language](#)) that acts as a container for one or more statements.

Generate Policy Start Over

Go back to **Edit Bucket Policy** and paste the JSON code in **Policy** window.

Click on **Save changes**.

Edit bucket policy

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

[Policy examples](#) [Policy generator](#)

Bucket ARN

arn:aws:s3:::bucketforpolicy258

Policy

```
1 {
2   "Id": "Policy1604558507423",
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "Stmnt1604558487902",
7       "Action": [
8         "s3:GetObject"
9       ],
10      "Effect": "Allow",
11      "Resource": "arn:aws:s3:::bucketforpolicy258/*",
12      "Condition": {
13        "IpAddress": {
14          "aws:SourceIp": "[redacted]"
15        }
16      },
17      "Principal": "*"
18    }
19  ]
20 }
```

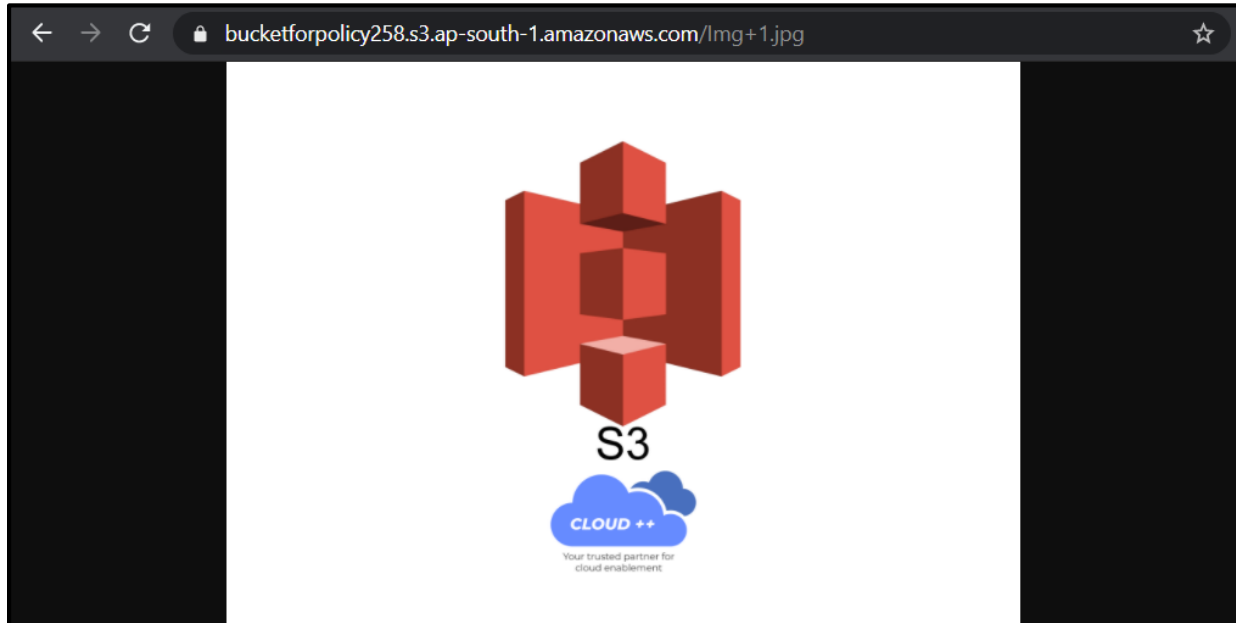
Cancel **Save changes**

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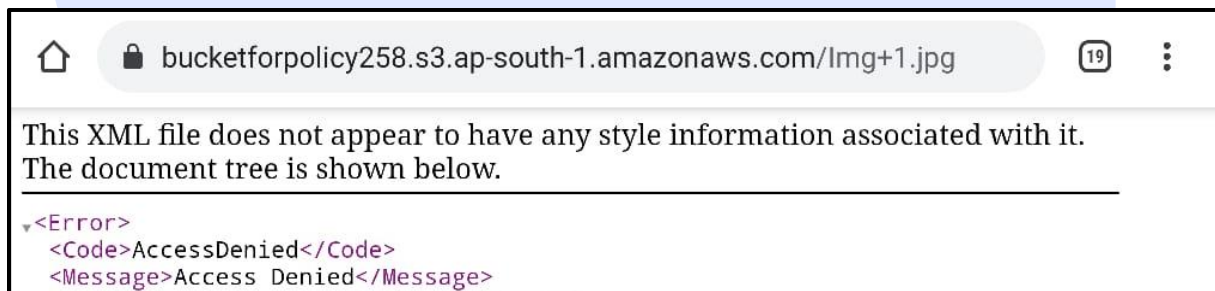


Step 4: Go back to bucket and test access from your IP address.

Click on any one of the images, copy the **URL**. Run the URL. This will display the image.

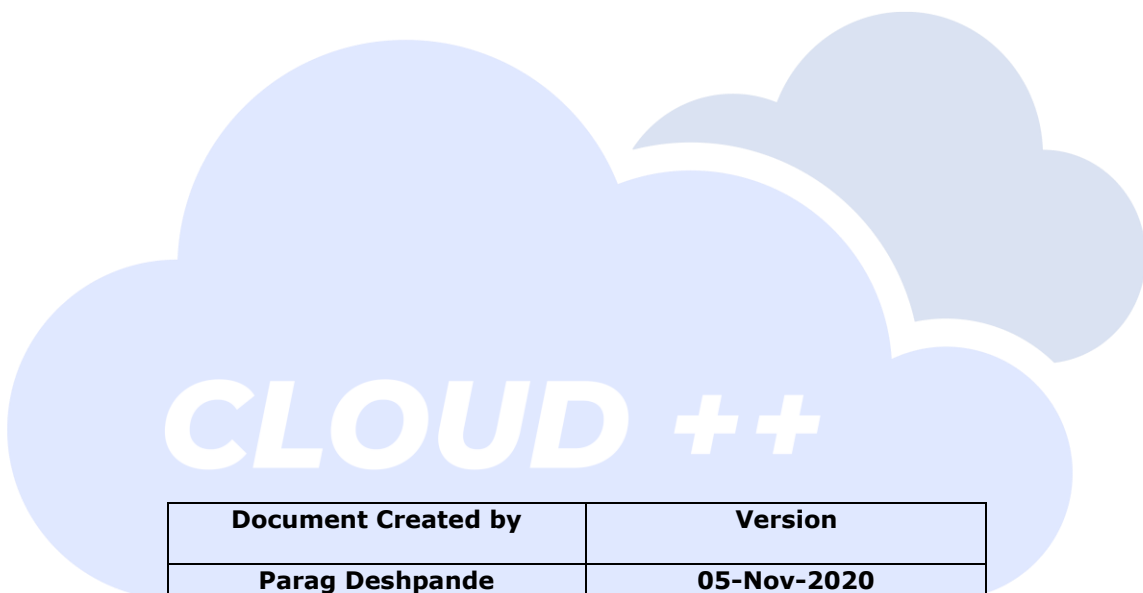


Run it using a **different network** (different Internet Service Provider). It would give an error.



Thus a specific IP access is established.

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Parag Deshpande	05-Nov-2020

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